

Nathan (Kaiyan) Zhang

Vancouver, BC | nathanzhang1017@gmail.com | LinkedIn | GitHub | Website

Education

The University of British Columbia

Sep 2023 – May 2028

B.A., Combined Major in Economics and Statistics, Minor in Data Science

Cumulative average: 93.5/100

Selected coursework: Statistical Inference (100), Machine Learning and Data Mining (93), Causal Methods (98)

Advanced Econometrics (98), Statistical Learning (97), Time Series and Forecasting (95)

Experience

ESG Co-op (Quantitative Research)

Sep 2026 – Present

British Columbia Investment Management Corporation (BCI) / Vancouver, BC

Capital Markets & Credit Investments

- Conduct quantitative analysis of company-level financial and ESG data to support fundamental investment research.
- Assess ESG risks and opportunities and summarize findings for the investment team.

Research Assistant

May 2026 – Present

UBC Sauder School of Business | Supervisor: Allen Hu

- Audited 1,330 RavenPack publication values against 3,210,045 ProQuest articles for coverage and data quality.
- Built media collection and validation for an 85,259-ID financial-creator directory, preserving raw files and provenance.

Research Assistant

Apr 2026 – Present

UBC Sauder School of Business | Supervisors: J. Frank Li and Cheng Chen

- Sole RA on a faculty-led study of age-biased technological change. Linked CPS, PSID, NLSY, O*NET, and PATSTAT to estimate wage and occupational sorting models.
- Harmonized 60+ O*NET releases and compared task text with about 56M PATSTAT records. Implemented GMM, a 999-replication block bootstrap, and an R replication package.

Research Assistant

Jan 2025 – May 2026

Vancouver School of Economics at UBC | Supervisors: Jonathan Graves and Laura Nelson

- Built Python/R modules and automated Quarto/GitHub Actions publishing for 1,000+ cumulative student users.
- Designed and taught a computer vision workshop. Coauthored an image-embedding notebook and a Programming Historian tutorial under site-maintainer review.

Selected Research

LLM Evaluation & Memory Efficiency

2026

Manuscript submitted, under review. Co-developed LLM memory evaluation with Jerrold Huang. In a separate 144-item benchmark, PSCC improved exact-match accuracy by 6.94 percentage points over A-Mem (95% CI: 2.64–11.32), using 67.2% as many API tokens.

Financial NLP & Return Prediction

2026

Manuscript submitted, under review. Built a deduplicated panel of about 31,000 firm events (2018–2025) with Jerrold Huang, linking Capital IQ transcripts, IBES, and CRSP. Tested whether management-answer novelty predicts returns beyond earnings surprise using time holdouts, placebo tests, and timing checks.

Causal Analysis of Consumer Spending

2025

Course research project. Applied cross-fitted DoubleML to 5,583 respondents and 22,332 in-person person-days. Under selection on observables and overlap, estimated a roughly 5% conditional effect of lower-friction payment mixes on daily spending.

Technical Skills

Programming & data: Python, R, SQL, Julia, Databricks, Bloomberg, DuckDB, Polars, Git/GitHub Actions, HPC/Slurm, Jupyter/Quarto

Modeling: causal inference/econometrics, panel/time series, machine learning, NLP, computer vision, scikit-learn, PyTorch, Transformers, fixest, statsmodels

Languages: Mandarin Chinese: native / English: academic working proficiency / Japanese: JLPT N3